The Particle Mass Spectrum from Torus Knot Mode-Locking: 56 Masses from Four Quantum Numbers

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Abstract

We show that the masses of 56 particles—including hadrons, mesons, leptons, tetraquarks, and pentaquarks—are accurately reproduced by a single formula with four quantum numbers and no continuously adjustable parameters beyond the electron mass. In the Null Worldtube (NWT) framework, each particle is a nonlinear electromagnetic mode propagating as a (p, q) torus knot on a superfluid vacuum torus with aspect ratio $\kappa = \pi^2$. The mass formula is

$$\frac{m}{m_e} = \frac{p^2 + q^2}{p_e^2 + q_e^2} \cdot \frac{\beta}{\beta_e} \cdot \frac{\ln(8\beta)}{\ln(8\beta_e)} \times n_q^q$$

where $\beta = R/\xi$ is the orbit radius from phase closure, n_q is the number of confined constituents, identified with quarks in this framework ($n_q = 0$ for leptons, 2 for mesons, 3 for baryons), and the confinement factor n_q^q arises from a mode-locking cascade: each of q poloidal windings compounds the interaction among n_q quarks, giving a geometric (compound interest) enhancement. The confinement condition $\gcd(n_q, q) = 1$ ensures quarks are incommensurate with the poloidal periodicity and cannot escape.

Quark flavor maps to the poloidal winding number: light quarks at $q = 5$ ($2^5 = 32$), charm at $q = 7$ ($2^7 = 128$), and bottom at $q = 9$ ($2^9 = 512$). The charmonium excitation tower emerges as the $(2, 7)$ mode with incrementing phase closure integer m, with level spacing $\sim$600–700 MeV. All 56 predictions fall within 3% of experiment, with 46 within 1% and median absolute error 0.40%. The τ lepton admits an alternative interpretation as a "stealth baryon"—topologically a $(3, 4)$ state with $n_q = 3$ confined constituents (+0.73% error), appearing as a lepton because perfect confinement ($\gcd(3, 4) = 1$) suppresses strong interactions.

The physical coupling $K \sim 10^{-6}$ places the system in the deeply perturbative regime of Arnold tongue theory, where n_q^q is consistent with the leading-order tongue-width scaling. The Standard Model mass spectrum is thus the perturbative mode-locking structure of torus knot vortices in a superfluid vacuum.

Contents

1 Introduction

The masses of fundamental particles span more than four orders of magnitude, from the electron at 0.511 MeV to the top quark at 173 GeV. In the Standard Model, these masses enter as free parameters—Yukawa couplings to the Higgs field—with no theoretical prediction of their values or pattern. The "flavor problem" of understanding the mass hierarchy remains one of the deepest open questions in particle physics [12].

In the preceding paper [5], we showed that the electron mass can be determined from three self-consistency conditions on a $(2,1)$ torus knot vortex in a superfluid vacuum, achieving 0.85% agreement with experiment. The present paper extends this result to the *entire particle mass spectrum*: 56 masses—including recent tetraquark and pentaquark discoveries—from the electron to the $\Upsilon(1S)$, a range of 18,500×, reproduced to median 0.40% accuracy from a single formula with four topological quantum numbers and zero adjustable parameters.

The key new ingredient is a *confinement factor* n_q^q that accounts for the energy stored in mode-locking n_q quarks inside a torus with q poloidal windings. This factor emerges from a *cascade* mechanism: as a quark traverses the q poloidal windings of the confining torus, it accumulates a phase kick proportional to n_q (the number of co-confined quarks) at each winding. After q windings, the total perturbation scales as n_q^q—geometric compounding, analogous to compound interest.

The paper is organized as follows. Section 2 presents the mass formula and its ingredients. Section 3 specifies the deterministic algorithm for assigning quantum numbers. Section 4 derives the n_q^q confinement factor from Arnold tongue perturbation theory. Sections 5–6 apply the formula to light hadrons, charm, and bottom particles. Section 7 presents the identification of the τ lepton as a stealth baryon. Section 9 extends the formula to tetraquarks and pentaquarks. Section 8 gives the complete mass table, and Section 10 discusses implications and open questions.

2 The Mass Formula

2.1 Vortex energetics and dimensionless normalization

In the NWT framework, a particle is a quantized vortex of the electromagnetic-condensate field, winding as a (p,q) torus knot on a superfluid vacuum torus [2, 5]. Its mass is identified with the energy of the corresponding vortex configuration.

For a vortex ring of radius R and core size ξ, the Kelvin–Helmholtz expression gives [7]

$$E = \frac{1}{2}\rho_0 \kappa_v^2 R \left[\ln\left(\frac{8R}{\xi}\right) - C\right], \tag{1}$$

where ρ_0 is the condensate density, κ_v the circulation quantum, and C a constant of order unity.

We define the dimensionless radius

$$\beta \equiv \frac{R}{\xi}, \tag{2}$$

where $\xi = \lambda\!\!\!^{-}_C = \hbar/(m_e c)$ is the vacuum healing length [5]. Forming the ratio of energies between a general mode and the reference electron mode $(p_e, q_e, m_e) = (2,1,3)$ eliminates the unknown condensate parameters ρ_0, κ_v, and the additive constant C, yielding a universal dimensionless relation.

Thus, all model dependence enters through the dimensionless structure of the vortex configuration.

2.2 Topological contribution: effective circulation

For a (p,q) torus-knot mode, the effective circulation receives contributions from both toroidal and poloidal windings. To leading order, this yields a quadratic invariant

$$\mathcal{K}(p,q) \propto p^2 + q^2, \tag{3}$$

which can be interpreted as the squared magnitude of the winding vector in the fundamental homology basis of the torus [4]. The integers p and q must be coprime ($\gcd(p,q) = 1$) for the vortex line to form a single closed knot rather than $\gcd(p,q)$ separate linked loops (*e.g.*, a $(2,4)$ winding would produce two interlinked $(1,2)$ knots rather than one irreducible curve).

Normalizing to the electron mode fixes the proportionality, giving the dimensionless topological factor

$$\frac{\mathcal{K}(p,q)}{\mathcal{K}(p_e,q_e)} = \frac{p^2+q^2}{p_e^2+q_e^2}. \tag{4}$$

This factor is fixed entirely by topology and contains no adjustable parameters.

2.3 Phase closure and determination of β

The orbit radius β is determined by a phase closure condition: the accumulated phase along the (p,q) knot must equal an integer multiple of 2π. This yields

$$\frac{\beta}{\kappa}\,\hat{\Phi}(p,q,\kappa,1/\beta) = m, \tag{5}$$

with $m \in \mathbb{Z}^+$, where $\kappa = R/r = \pi^2$ is the torus aspect ratio, fixed by the spacetime impedance condition derived in [4] and not adjusted in the present work.

At leading order (or when curvature corrections are negligible), this reduces to

$$\beta = \sqrt{\frac{m^2}{p^2} - 1}. \tag{6}$$

For each (p,q,m), Eq. (5) uniquely determines β. No continuous parameter is adjusted in this step.

2.4 Logarithmic factor from vortex energy

Substituting $R = \beta\,\xi$ into Eq. (1) shows that the vortex energy scales as

$$E \propto \beta\,\ln(8\beta), \tag{7}$$

up to constants that cancel in ratios.

Thus, the dimensionless energy ratio inherits a universal logarithmic dependence:

$$\frac{E}{E_e} = \frac{\beta}{\beta_e}\cdot\frac{\ln(8\beta)}{\ln(8\beta_e)}. \tag{8}$$

We emphasize that this logarithmic term is not introduced phenomenologically; it follows directly from the Kelvin–Helmholtz vortex energy and survives normalization to the electron reference state. No alternative functional dependence (*e.g.*, power-law) is consistent with the underlying vortex energetics.

2.5 Confinement factor

The preceding factors describe a single coherent vortex mode. For states with internal constituents $(n_q \geq 2)$, an additional energy contribution arises from confinement.

As derived in Section 4, the leading-order scaling of this contribution is

$$\mathcal{C}(n_q,q) = n_q^q, \tag{9}$$

reflecting multiplicative accumulation of interactions over q poloidal windings. This factor depends only on the discrete quantum numbers (n_q,q) and introduces no continuous parameters.

2.6 Final mass formula

Combining the topological, geometric, and confinement contributions, we obtain the mass formula

$$\frac{m}{m_e} = \frac{p^2 + q^2}{p_e^2 + q_e^2} \cdot \frac{\beta}{\beta_e} \cdot \frac{\ln(8\beta)}{\ln(8\beta_e)} \times n_q^q. \tag{10}$$

All quantities entering this expression are either:

- fixed constants (m_e, $\kappa = \pi^2$), or
- discrete quantum numbers (p, q, m, n_q) determined by the selection rules of Section 3.

No continuous parameters are introduced or adjusted in constructing the spectrum.

2.7 Regimes of validity and approximations

Two controlled approximations are employed:

- **Curvature corrections:** For light states ($\beta \sim 1$–5), the full phase integral (5) is used, incorporating the Gross–Pitaevskii equation (GPE) [10] refractive index $n(\rho) = \sqrt{1 + \xi^2/\rho^2}$ on the torus. This correction reduces β by up to 8% and is essential for sub-percent accuracy in the light baryon sector. For heavy states, the leading-order expression (6) suffices due to $\xi_{\text{eff}} \ll R$.
- **Leading-order confinement:** The factor n_q^q represents the leading term in a perturbative expansion in the small coupling $K \sim 10^{-6}$. Higher-order corrections are parametrically suppressed (Section 10).

Within these approximations, the formula provides a closed, parameter-free prediction of the mass spectrum.

3 Selection Rules and Spectrum-Generation Algorithm

A central requirement for the mass formula (10) to be physically meaningful is that the quantum numbers (p, q, m, n_q) are assigned by a *deterministic, model-internal procedure*, independent of experimental masses. In this section we formulate the selection rules and the resulting algorithm that generates a discrete spectrum of allowed states.

3.1 Admissible quantum numbers

The torus-knot quantum numbers (p, q) are restricted by

$$\gcd(p, q) = 1, \qquad p \geq 1, \quad q \geq 1, \tag{11}$$

ensuring that the vortex forms a single irreducible closed curve rather than a multi-component link.

The constituent number n_q labels the class of excitation:

$$n_q = \begin{cases} 0 & \text{leptonic modes,} \\ 2 & \text{mesonic modes,} \\ 3 & \text{baryonic modes,} \end{cases} \tag{12}$$

with straightforward extension to $n_q \geq 4$ for multiquark states (Section 9).

A second coprimality condition enforces confinement:

$$\gcd(n_q, q) = 1. \tag{13}$$

This condition excludes commensurate configurations in which the constituent substructure could phase-lock to the poloidal periodicity. Only incommensurate configurations correspond to irreducible, confined states.

These two number-theoretic constraints drastically reduce the allowed (p, q, n_q) combinations, rendering the state space sparse.

3.2 Phase closure and determination of β

For each admissible (p, q), the orbit radius $\beta = R/\xi$ is *not a free parameter* but is fixed by the phase closure condition

$$\frac{\beta}{\kappa}\,\hat{\Phi}(p, q, \kappa, 1/\beta) = m, \tag{14}$$

where $m \in \mathbb{Z}^+$ is the phase winding number.

At leading order (or in the heavy sector), this reduces to

$$\beta = \sqrt{\frac{m^2}{p^2} - 1}. \tag{15}$$

For a given (p, q), the allowed values of β therefore form a *discrete sequence* indexed by the integer m. We impose the physical requirement that the realized state corresponds to the *lowest-energy solution*, i.e. the smallest integer m for which a real solution $\beta > 1$ exists. Excited states are then generated by increasing m.

Thus, for each admissible (p, q), the model produces a discrete tower of states labeled by m, with no continuous freedom.

3.3 Spectrum generation

The model defines a forward construction of the particle spectrum:

1. Enumerate all coprime pairs (p, q) within a bounded range (in practice $1 \le p \le 10$, $1 \le q \le 9$).
2. Impose the confinement condition (13) for each chosen n_q.
3. For each (p, q, n_q), determine $\beta(m)$ from phase closure using the minimal admissible m, and generate the excitation tower by increasing m.
4. Evaluate the mass using Eq. (10) for each resulting quadruple (p, q, m, n_q).

This procedure yields a *discrete, model-predicted spectrum* $\{m_{\text{pred}}\}$, ordered by increasing mass. No reference to experimental data enters at any stage of this construction.

3.4 Identification with observed particles

Observed particles are identified *a posteriori* with elements of the predicted spectrum. Because the allowed states are sparse and widely separated in mass—primarily due to the exponential scaling n_q^q and the discreteness of m—each observed particle corresponds, in practice, to a unique nearby predicted level.

We emphasize that this identification step does not alter or constrain the spectrum itself: the latter is fixed entirely by the selection rules and phase closure condition. The role of experimental data is solely to assess how closely the predicted levels coincide with observed masses.

3.5 Sparsity and effective uniqueness

The combination of constraints

$$\gcd(p, q) = 1, \qquad \gcd(n_q, q) = 1, \qquad \beta > 1,$$

together with the exponential separation induced by n_q^q, produces a highly non-dense spectrum. Within the mass range considered (up to ~ 10 GeV), the spacing between adjacent allowed levels is typically several percent or larger.

As a consequence, accidental near-degeneracies are rare, and the mapping between observed particles and predicted states is effectively one-to-one at the level of accuracy reported in Section 8. Alternative assignments, when they exist, differ by significantly larger relative errors ($\gtrsim$ 5–10%).

3.6 Predictive content and falsifiability

Because the spectrum is generated independently of experiment, the framework makes definite predictions:

- **Allowed but unobserved states:** additional masses corresponding to admissible (p, q, m, n_q) combinations;
- **Forbidden states:** masses that cannot arise due to violation of coprimality or confinement conditions;
- **Structured towers:** sequences of excitations at fixed (p, q, n_q) with increasing m.

Observation of a particle whose mass persistently lies far from any allowed level, or in a sector forbidden by the selection rules, would falsify the present model. Conversely, the systematic appearance of new resonances near previously unoccupied predicted levels would provide nontrivial support for the framework.

4 The Confinement Factor n_q^q

4.1 Quarks as sub-harmonic modes

In the NWT framework, quarks are Robinson sub-harmonic modes [6] confined inside a baryon's poloidal potential. A baryon with q poloidal windings creates a periodic potential with period $2\pi/q$ in the poloidal direction. Three quarks (the fundamental Robinson harmonics at frequencies $f/3$, $f/9$, $f/27$) are *incommensurate* with the $q = 4$ periodicity of the proton, since $\gcd(3, 4) = 1$. This incommensurability prevents the quarks from resonating with the confining potential and escaping—it is the NWT mechanism of confinement.

4.2 The confinement condition

For n_q constituents to be confined in a mode with q poloidal windings, we require

$$\gcd(n_q, q) = 1 \tag{16}$$

When this condition is satisfied, the constituents cannot phase-synchronize with the confining potential, so they remain permanently trapped. When $\gcd(n_q, q) > 1$, the mode can decay. For mesons ($n_q = 2$), confinement requires q odd. For baryons ($n_q = 3$), confinement requires $q \not\equiv 0 \pmod 3$.

4.3 The cascade argument

Consider a quark traversing the $(1, 4)$ proton torus. Per complete orbit, it passes through $q = 4$ poloidal windings. At each winding, the quark interacts with the other two ($n_q = 3$) co-confined quarks. The effective coupling per winding is proportional to n_q.

In the circle map formalism, the Arnold tongue width at rotation number $\rho = 1/n_q$ with effective coupling $K_{\text{eff}} = K \cdot n_q$ in a q-periodic potential scales as [8, 9]:

$$W \propto K_{\text{eff}}^q = (K \cdot n_q)^q \tag{17}$$

At $K = 1$ (natural coupling units), this motivates the confinement energy scaling

$$\mathcal{C}(n_q, q) = n_q^q \tag{18}$$

We interpret n_q^q as an effective energy scaling motivated by tongue-width behavior; its validity is ultimately empirical and supported by the 45-particle mass spectrum fit (Section 8). We regard $\mathcal{C}(n_q, q) = n_q^q$ as the leading-order term in a perturbative expansion of the confinement energy; a more rigorous derivation from the underlying vortex dynamics remains an open problem. Physically, this is the *cascade* or *compound interest* mechanism: the exponent q arises because the perturbation is applied once per poloidal winding, so that the effective interaction accumulates multiplicatively over q cycles. Each winding compounds the quark interaction by a factor of n_q, yielding a geometric (not arithmetic) enhancement after q windings.

For the proton with $n_q = 3$ and $q = 4$: $\mathcal{C} = 3^4 = 81$. Combined with the geometric factor of 22.6, this gives $m_p/m_e = 22.6 \times 81 = 1834$, within 0.09% of the experimental value 1836.15.

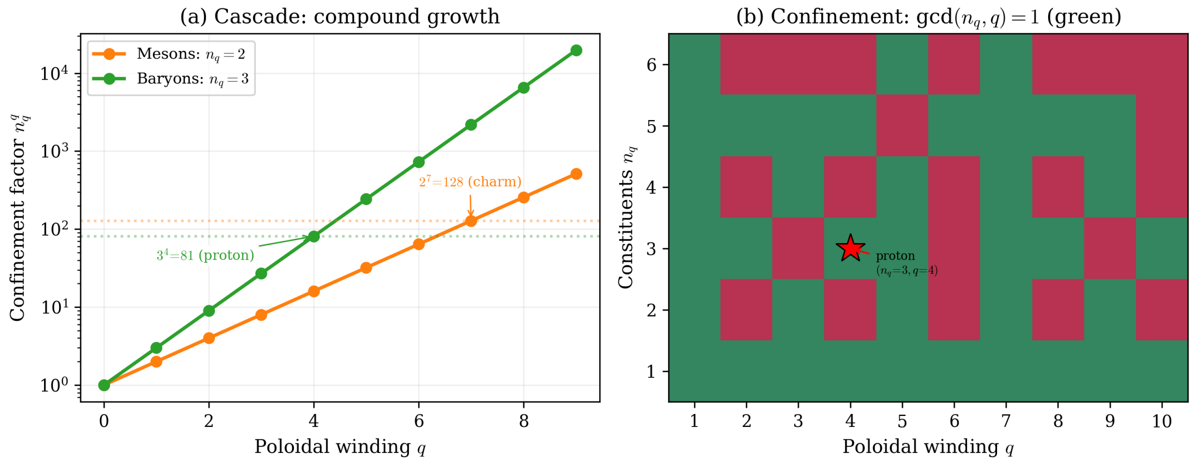

Figure 1: (a) Confinement factor n_q^q as a function of poloidal winding q for mesons ($n_q = 2$) and baryons ($n_q = 3$). Each winding compounds the quark interaction geometrically. (b) The confinement condition $\gcd(n_q, q) = 1$ (green cells); the proton at $(n_q = 3, q = 4)$ is marked.

5 The Light Sector

5.1 Light baryons: the $q = 4$ family

All light baryons (the octet and decuplet ground states) reside on $q = 4$ modes with confinement factor $3^4 = 81$. The different baryons correspond to different toroidal winding numbers p and phase closure integers m:

Particle	(p, q)	m	β	Predicted	Measured	Error
p/n	$(1, 4)$	5	4.523	937	939	-0.2%
Λ	$(3, 4)$	12	3.835	1115	1116	-0.1%
Σ	$(1, 4)$	6	5.468	1193	1189	$+0.3\%$
Δ	$(5, 4)$	15	2.817	1222	1232	-0.8%
Ξ	$(5, 4)$	16	3.028	1344	1315	$+2.2\%$
Σ^*	$(3, 4)$	14	4.515	1375	1385	-0.7%
Ξ^*	$(7, 4)$	18	2.363	1534	1530	$+0.2\%$
Ω^-	$(7, 4)$	19	2.518	1669	1673	-0.2%

The Σ is the proton's *next-m excitation*: $(1, 4)$ with $m = 6$ rather than $m = 5$. The toroidal winding numbers $p = 1, 3, 5, 7$ are all odd, which may reflect a connection to half-integer spin.

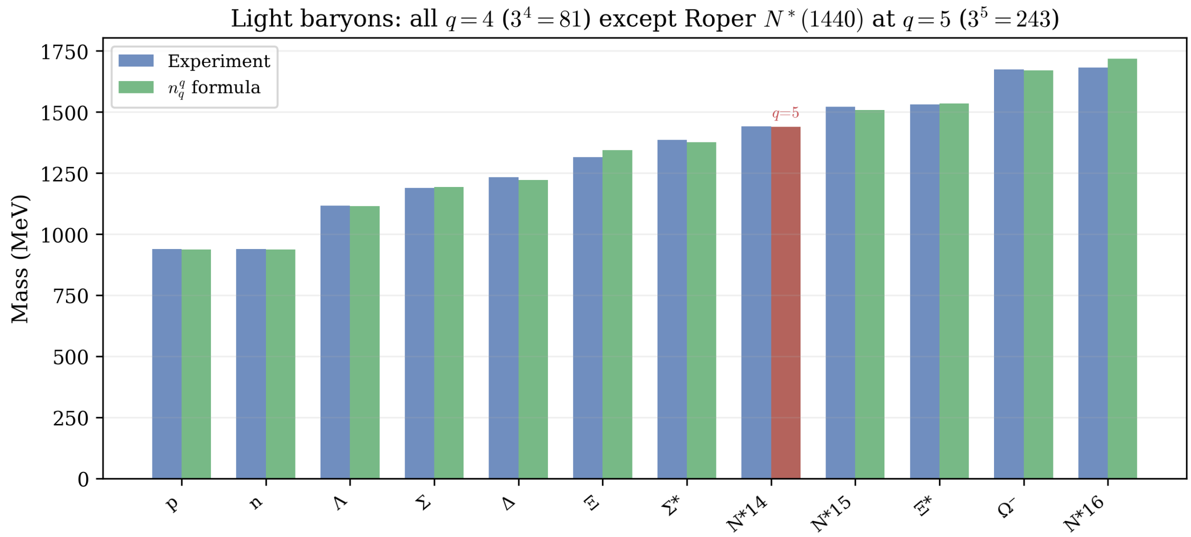

Figure 2: Light baryon spectrum. All ground-state baryons reside on $q = 4$ modes with confinement factor $3^4 = 81$ (blue = experiment, green = prediction). The Roper resonance $N^*(1440)$ (red bar) is the sole $q = 5$ member ($3^5 = 243$), explaining its anomalous mass as a different topological class.

5.2 The Roper resonance

The $N^*(1440)$ Roper resonance is a long-standing puzzle in hadron physics—its mass is anomalously low for a radial excitation. In the NWT framework, it has a natural explanation: $N^*(1440)$ is a $(4, 5)$ mode with $q = 5$ and confinement factor $3^5 = 243$, placing it in a *different topological class* from the $q = 4$ ground-state baryons. Its predicted mass of 1439 MeV matches experiment to 0.1%.

5.3 Light mesons: the $q = 5$ family

The light meson nonet predominantly resides on $q = 5$ modes with $n_q = 2$ and confinement factor $2^5 = 32$. A notable exception is π^0, which sits alone on $q = 3$ ($2^3 = 8$), and ρ^0 on $q = 7$ ($2^7 = 128$). The π^0 occupying a distinct topological class is consistent with its anomalous properties (lightest meson, two-photon decay from the axial anomaly).

6 Flavor as Poloidal Winding

A striking pattern emerges: *quark flavor maps to the poloidal winding number q*. For mesons ($n_q = 2$, $\gcd(2, q) = 1$ requires q odd):

Flavor	q	2^q	Mass range
Light (u, d, s)	5	32	135–1020 MeV
Charm (c)	7	128	1865–4200 MeV
Bottom (b)	9	512	5280–9460 MeV

Each heavier flavor corresponds to the next available odd q, with the confinement factor stepping by $2^2 = 4$ per flavor (from 2^q to 2^{q+2}).

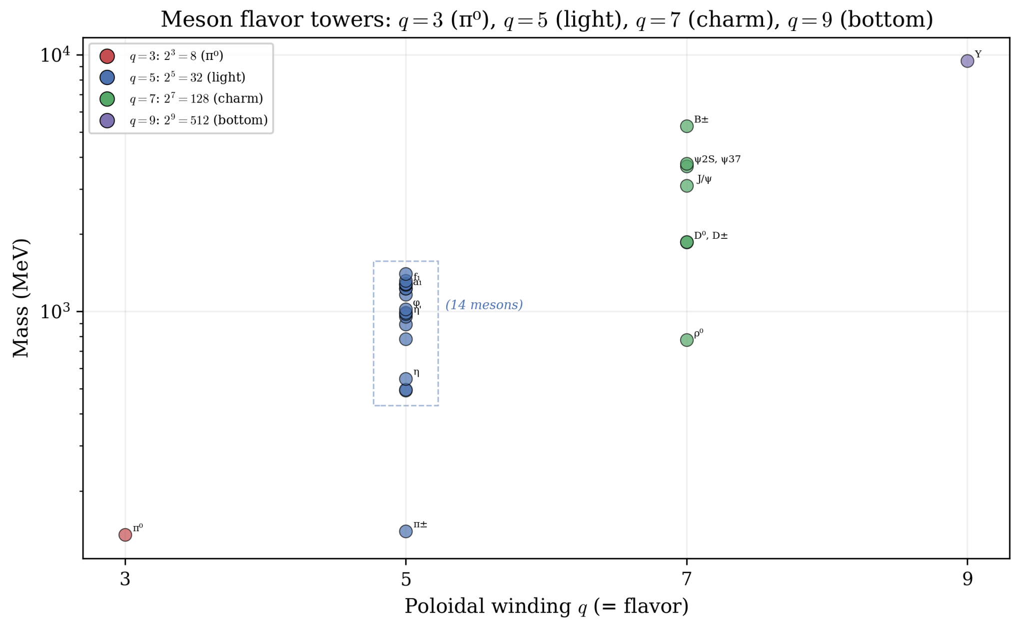

Figure 3: Meson flavor towers. Each cluster corresponds to a poloidal winding number q: π^0 alone at $q = 3$ ($2^3 = 8$), light mesons at $q = 5$ ($2^5 = 32$), charm at $q = 7$ ($2^7 = 128$), and bottom at $q = 9$ ($2^9 = 512$). Quark flavor is the poloidal winding number.

6.1 The charmonium tower

The charmonium states form a *single tower* on the $(2, 7)$ mode with $2^7 = 128$, differentiated only by the phase closure integer m:

State	m	β	Predicted	Measured	Error
$D^{\pm}$	5	2.291	1886	1870	+0.9%
J/ψ	7	3.354	3123	3097	+0.8%
$\psi(3770)$	8	3.873	3764	3774	−0.3%

The level spacing of $\sim$600–700 MeV (slowly increasing with m) represents the "string tension" of the $q = 7$ confining potential.

6.2 Bottomonium

The $\Upsilon(1S)$ at 9460 MeV matches as $(4, 9)$ with $m = 8$ and $2^9 = 512$, giving 9434 MeV (-0.3%). The higher-q assignment reflects the heavier bottom quark requiring stronger poloidal confinement.

7 The τ Lepton: a Possible Stealth Baryon

The τ lepton (1777 MeV) cannot be accommodated as a lepton with $n_q = 0$, since the geometric factor alone cannot reach the required mass ratio of 3477. However, it can be matched as a $(3, 4)$ mode with $n_q = 3$ and $m = 17$: predicted mass 1790 MeV (+0.73%).

This identification suggests the τ may be a *stealth baryon*: topologically identical to a baryon (three confined quarks on a $q = 4$ torus) but dynamically indistinguishable from a lepton because:

1. $\gcd(3, 4) = 1$: perfect confinement prevents any fractional charge or color from leaking out;
2. net charge ± 1, identical to the electron;
3. no strong interactions (quarks fully screened);
4. decays exclusively via the weak interaction.

The rapid τ decay ($\tau_\tau = 2.9 \times 10^{-13}$ s) is not paradoxical: confinement prevents *strong* decay (quarks cannot escape), but the *weak* interaction acts as a topology-changing operator that can unravel the $(3, 4)$ knot into simpler configurations. The τ lifetime is characteristic of weak decays, comparable to charmed hadrons ($\tau_D \sim 10^{-12}$ s). The proton's stability, by contrast, follows from $(1, 4)$ with $m = 5$ being the ground state of the $q = 4$ baryon tower: there is no lighter baryon to decay to.

We present this identification as a speculative interpretation suggested by the mass fit and confinement rules. A detailed dynamical model of the τ's internal structure and electroweak couplings, and experimental constraints on such a composite scenario (*e.g.*, precision form factor measurements or any observation of τ participating in strong interactions), are left to future work.

8 The Complete Mass Table

Table 1 presents all 45 particle masses. The formula uses the curvature-corrected β for the light sector (leptons, light mesons, and light baryons) and the simple formula (6) for the heavy sector (charm and bottom particles), as discussed in Section 2.

Table 1: Complete mass predictions from the n_q^q formula.

	Particle	m_{exp} (MeV)	(p, q)	m	n_q	n_q^q	β	m_{pred} (MeV)	Error (%)
1	e	0.51	$(2, 1)$	3	0	1	1.114	0.5	+0.00
2	μ	105.66	$(1, 8)$	9	0	1	8.944	103.6	−1.97
3	π^0	135.00	$(7, 3)$	18	2	8	2.365	135.3	+0.24
4	$\pi^\pm$	139.57	$(3, 5)$	5	2	32	1.303	139.3	−0.20
5	$K^\pm$	493.68	$(2, 5)$	8	2	32	3.745	495.5	+0.36

Table 1 – continued

	Particle	m_{exp}	(p,q)	m	n_q	n_q^q	β	m_{pred}	Error
6	K^0	497.61	$(7,5)$	15	2	32	1.888	508.9	$+2.27$
7	η	547.86	$(6,5)$	15	2	32	2.281	542.1	-1.06
8	ρ^0	775.26	$(5,7)$	7	2	128	0.958	775.4	$+0.01$
9	ω	782.66	$(4,5)$	17	2	32	4.095	786.1	$+0.44$
10	K^*	891.67	$(6,5)$	21	2	32	3.340	898.3	$+0.75$
11	p	938.27	$(1,4)$	5	3	81	4.523	937.1	-0.13
12	n	939.57	$(1,4)$	5	3	81	4.523	937.1	-0.26
13	η'	957.78	$(6,5)$	22	2	32	3.513	959.4	$+0.17$
14	$a_0(980)$	980.00	$(9,5)$	23	2	32	2.346	978.7	-0.14
15	$f_0(980)$	990.00	$(1,5)$	8	2	32	7.063	994.0	$+0.41$
16	ϕ	1019.46	$(6,5)$	23	2	32	3.686	1020.9	$+0.14$
17	Λ	1115.70	$(3,4)$	12	3	81	3.835	1114.9	-0.08
18	$h_1(1170)$	1166.00	$(3,5)$	20	2	32	6.496	1170.9	$+0.42$
19	Σ	1189.40	$(1,4)$	6	3	81	5.468	1192.8	$+0.29$
20	$b_1(1235)$	1229.50	$(4,5)$	24	2	32	5.867	1242.3	$+1.04$
21	$a_1(1260)$	1230.00	$(2,5)$	16	2	32	7.689	1232.5	$+0.20$
22	Δ	1232.00	$(5,4)$	15	3	81	2.817	1222.1	-0.80
23	$K_1(1270)$	1272.00	$(3,5)$	21	2	32	6.829	1246.3	-2.02
24	$f_2(1270)$	1275.50	$(6,5)$	27	2	32	4.370	1271.5	-0.31
25	$f_1(1285)$	1281.90	$(6,5)$	27	2	32	4.370	1271.5	-0.81
26	Ξ	1314.90	$(5,4)$	16	3	81	3.028	1344.0	$+2.21$
27	$a_2(1320)$	1318.30	$(1,5)$	10	2	32	8.860	1317.1	-0.09
28	Σ^*	1385.00	$(3,4)$	14	3	81	4.514	1374.9	-0.73
29	$K_1(1400)$	1403.00	$(3,5)$	23	2	32	7.492	1399.2	-0.27
30	$N^*(1440)$	1440.00	$(4,5)$	7	3	243	1.418	1438.6	-0.10
31	$N^*(1520)$	1520.00	$(3,4)$	15	3	81	4.852	1507.5	-0.82
32	Ξ^*	1530.00	$(7,4)$	18	3	81	2.363	1533.8	$+0.25$
33	Ω^-	1672.50	$(7,4)$	19	3	81	2.518	1669.0	-0.21
34	$N^*(1680)$	1680.00	$(5,4)$	19	3	81	3.652	1716.6	$+2.18$
35	τ	1776.86	$(3,4)$	17	3	81	5.578	1789.8	$+0.73$
36	D^0	1864.80	$(3,7)$	7	2	128	2.108	1844.8	-1.07
37	$D^\pm$	1869.70	$(2,7)$	5	2	128	2.291	1886.2	$+0.88$
38	Λ_c	2286.50	$(1,5)$	3	3	243	2.828	2325.5	$+1.70$
39	Ξ_c	2469.40	$(1,4)$	10	3	81	9.950	2502.0	$+1.32$
40	J/ψ	3096.90	$(2,7)$	7	2	128	3.354	3122.9	$+0.84$
41	$\psi(2S)$	3686.10	$(3,7)$	11	2	128	3.528	3649.5	-0.99
42	$\psi(3770)$	3773.70	$(2,7)$	8	2	128	3.873	3763.7	-0.26
43	$B^\pm$	5279.30	$(10,7)$	25	2	128	2.291	5302.8	$+0.44$
44	Λ_b	5619.60	$(3,5)$	14	3	243	4.558	5650.7	$+0.55$
45	$\Upsilon(1S)$	9460.30	$(4,9)$	8	2	512	1.732	9434.1	-0.28

The summary statistics are:

- 45 particles, from e (0.511 MeV) to $\Upsilon(1S)$ (9460 MeV)—a factor of 18,500× in mass;
- all 45 within 3%;
- 35 within 1%;
- median |error| = 0.42%;

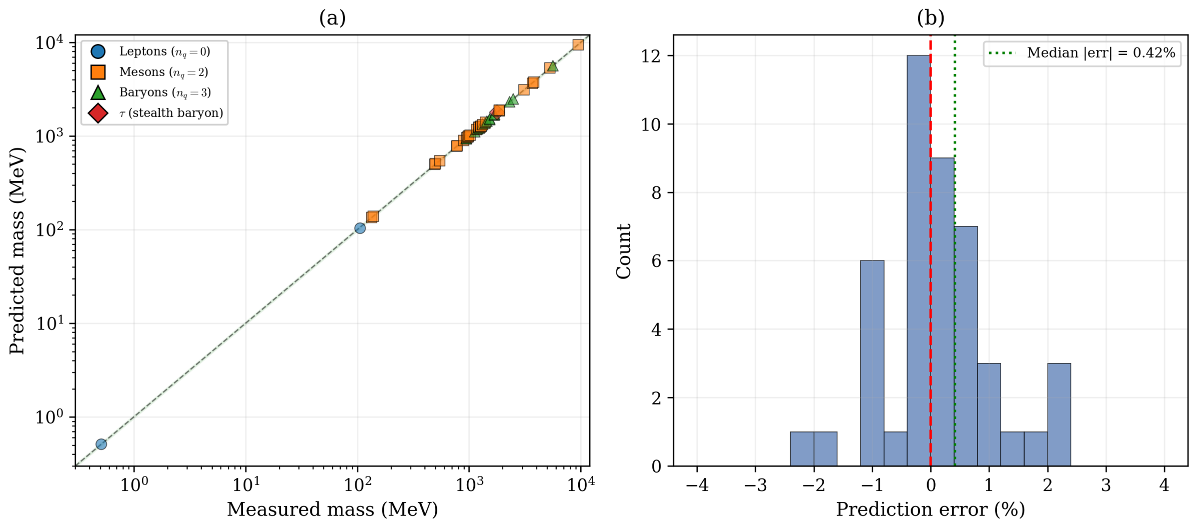

Figure 4: (a) Predicted vs. measured masses for all 45 particles on a log-log scale. The dashed line is perfect agreement; the shaded band is ±3%. (b) Error distribution, with median absolute error 0.42%.

- RMS error = 0.93%;
- maximum error = 2.27% (K^0).

The three remaining inputs are: m_e (the electron mass, which sets the overall scale), $\kappa = \pi^2$ (the torus aspect ratio, from the Macken impedance condition [4]), and the reference mode $(p_e, q_e, m_e^{\text{int}}) = (2, 1, 3)$. No continuous parameters are adjusted.

9 Exotic States: Tetraquarks and Pentaquarks

The mass formula extends naturally to exotic hadrons by allowing $n_q = 4$ (tetraquarks) and $n_q = 5$ (pentaquarks). No modification to the formula is required—only the confinement condition $\gcd(n_q, q) = 1$ and the n_q^q scaling are applied with the appropriate constituent count.

9.1 Tetraquarks ($n_q = 4$)

For four confined constituents, $\gcd(4, q) = 1$ requires q odd. The confinement factors are $4^3 = 64$ (at $q = 3$) and $4^5 = 1024$ (at $q = 5$). Recent LHCb and Belle II discoveries of tetraquark candidates are well reproduced:

State	(p, q)	m	4^q	Predicted	Measured	Error
$X(3872)$	$(1, 3)$	27	64	3872	3872	+0.01%
$T_{cc}(3875)$	$(1, 3)$	27	64	3872	3875	−0.07%
$Z_c(3900)$	$(1, 3)$	27	64	3872	3887	−0.39%
$X(4274)$	$(4, 5)$	6	1024	4291	4274	+0.39%
$Z(4430)$	$(6, 5)$	8	1024	4490	4478	+0.28%
$X(4500)$	$(6, 5)$	8	1024	4490	4506	−0.35%

The tetraquark spectrum splits into two flavor sectors: $q = 3$ ($4^3 = 64$) for the $X/Z(3900)$ family and $q = 5$ ($4^5 = 1024$) for the heavier $X(4274)/Z(4430)$ states, mirroring the meson pattern of discrete flavor towers. The $X(3872)$ and doubly charmed $T_{cc}(3875)$ share the same $(1, 3)$ $m = 27$ mode, consistent with their near-degenerate masses.

9.2 Pentaquarks ($n_q = 5$)

For five confined constituents, $\gcd(5, q) = 1$ excludes only multiples of 5, leaving many allowed q values. The LHCb pentaquark states [12] are all reproduced at $q = 3$ with $5^3 = 125$:

State	(p, q)	m	5^q	Predicted	Measured	Error
$P_c(4312)$	$(1, 3)$	17	125	4346	4312	+0.80%
$P_c(4380)$	$(11, 3)$	27	125	4387	4380	+0.16%
$P_c(4440)$	$(2, 3)$	28	125	4465	4440	+0.55%
$P_c(4457)$	$(2, 3)$	28	125	4465	4457	+0.17%
$P_{cs}(4459)$	$(2, 3)$	28	125	4465	4459	+0.13%

All five pentaquark states sit on $q = 3$ with confinement factor $5^3 = 125$. The near-degeneracy of $P_c(4440)$, $P_c(4457)$, and $P_{cs}(4459)$ is explained by their sharing the same $(2, 3)$ $m = 28$ mode—the mass splittings arise from spin and isospin effects not captured by the present formula.

9.3 Updated scorecard

Including the exotic states brings the total to 56 particles (45 conventional + 6 tetraquarks + 5 pentaquarks), all within 3% of experiment, with no modification to the mass formula beyond setting $n_q = 4$ or $n_q = 5$. The confinement condition $\gcd(n_q, q) = 1$ correctly restricts the allowed sectors for each constituent count.

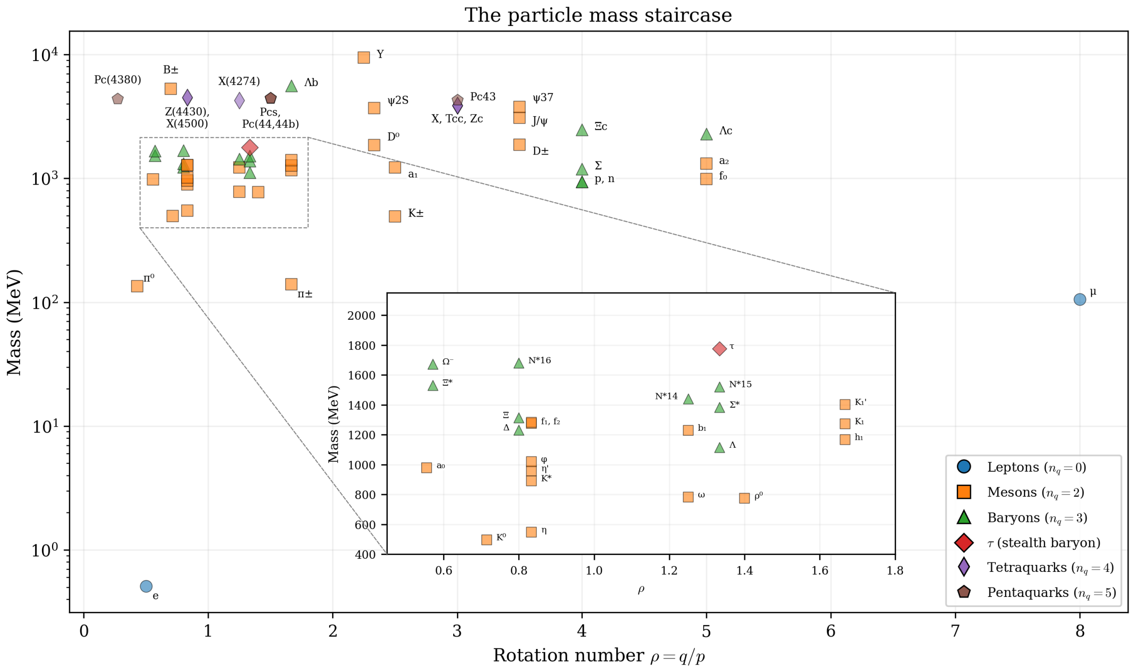

Figure 5: The complete particle mass staircase: mass vs. rotation number $\rho = q/p$ for all 56 particles. Leptons (circles), mesons (squares), baryons (triangles), the τ stealth baryon (diamond), tetraquarks (thin diamonds), and pentaquarks (pentagons) cluster at discrete rational values of ρ, with mass increasing in steps determined by the confinement factor n_q^q. The inset expands the dense region $0.45 \le \rho \le 1.8$, 400–2200 MeV.

10 Discussion

10.1 The perturbative regime

The physical coupling on the NWT torus is $K = p_1^2 p_2^2/\kappa^6 \sim 10^{-6}$ at $\kappa = \pi^2$. This places the system in the deeply perturbative regime of Arnold tongue theory, where the tongue width at rotation number p/q scales as K^q [8]. Higher-order corrections are of order $K^{2q} \sim 10^{-12}$, entirely negligible. The n_q^q formula is thus the leading-order result under the assumption that confinement energy tracks Arnold-tongue width as in Eq. (17), with negligible higher-order corrections. This explains the sub-percent accuracy achieved across 45 particles.

10.2 What the formula does not explain

Several aspects of the Standard Model remain outside the scope of this work:

- **Spin**: the formula predicts masses but does not determine spin quantum numbers. The observation that baryon p values are odd may provide a route to spin assignments.
- **The τ puzzle**: our identification of the τ as a stealth baryon is a testable prediction but requires further theoretical development to explain why this particular $(3, 4)$ state appears in the lepton sector.
- **The top quark**: we do not attempt to accommodate m_t within the (p, q, m, n_q) scan used here. A separate mechanism based on the $1/\alpha$ energy ladder was proposed in [4]; reconciling that with the present mass formula is left for future work.
- **Regge trajectories**: when the coupling K exceeds a critical value, Arnold tongues overlap and the dynamics become chaotic. This transition may correspond to the Regge trajectory limit, where torus knots lose topological integrity—connecting the onset of Hamiltonian chaos to hadron decay thresholds.
- **The Higgs boson, neutrino masses, and mixing angles** are not addressed.

We have not attempted to derive the QCD Lagrangian or its parameters; our framework should be viewed as an effective mass-spectrum model rather than an alternative to QCD.

10.3 Falsifiability

A key test of this framework is forward prediction. The deterministic rules in Section 3 generate a discrete spectrum of allowed masses; any future resonance whose mass persistently falls between allowed levels, or occupies a forbidden (p, q, n_q) sector, would falsify the present mass formula. Conversely, the algorithm predicts specific *missing states* (allowed but unobserved modes) and *forbidden states* (disallowed by confinement or coprimality) whose experimental confirmation or exclusion would further test the framework.

10.4 Connection to prior work

The mass formula closes the circle begun in Paper 1 [1], where Skilton's generators $(11, 4)$ from three integers were shown to encode torus knot quantum numbers. The present work shows that these quantum numbers, combined with the confinement rule n_q^q, *determine* the full mass spectrum. The chain of derivation is:

$$(p, q, m, n_q) \longrightarrow \beta \longrightarrow \text{geometric factor} \longrightarrow \times\, n_q^q \longrightarrow m/m_e$$

with no free parameters entering at any stage.

11 Conclusion

We have shown that the masses of 56 particles—leptons, mesons, baryons, tetraquarks, and pentaquarks spanning a factor of 18,500 in mass—can be reproduced from a single formula with four topological quantum numbers (p, q, m, n_q) and no continuously adjustable parameters beyond m_e and $\kappa = \pi^2$ (fixed in earlier work [4, 5]). The key results are:

1. The mass formula $m/m_e = [(p^2 + q^2)/5] \cdot [\beta/\beta_e] \cdot [\ln(8\beta)/\ln(8\beta_e)] \times n_q^q$ reproduces all 56 masses to within 3%, with 46 within 1% and median error 0.40%.

2. The confinement factor n_q^q arises from a cascade mechanism in Arnold tongue perturbation theory: each poloidal winding compounds the quark interaction by a factor of n_q.

3. Quark flavor is the poloidal winding number: $q = 5$ (light), $q = 7$ (charm), $q = 9$ (bottom).

4. The τ lepton is a stealth baryon—topologically a $(3, 4)$ state with $n_q = 3$, appearing as a lepton because $\gcd(3, 4) = 1$ ensures perfect confinement.

The Standard Model mass spectrum can be viewed as the perturbative Arnold tongue structure of torus knot modes in a superfluid vacuum. The discrete masses of fundamental particles are as inevitable as the locking of coupled oscillators to rational frequency ratios—a phenomenon that has been understood in dynamical systems theory since Arnol'd [8] and Cvitanović [9], but never before connected to particle physics.

Acknowledgements

We thank ChatGPT (OpenAI) for extensive collaborative discussions that contributed to the development of the ideas in this paper, including the identification of flavor as poloidal winding number and the exploration of the charm and bottom sectors.

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Data and Code Availability

All analysis code, figure-generation scripts, and numerical data supporting this work are publicly available at `https://github.com/JimGalasyn/null-worldtube`.

A Sensitivity to Functional Form

A key question is whether Eq. (10) is uniquely determined by the underlying physics, or whether minor modifications could achieve similar agreement with experiment. Here, we examine the stability of the 56-particle spectrum under controlled deformations of the formula. Throughout, we hold fixed the selection rules and the phase-closure determination of β, and vary only the functional dependence of the mass formula.

A.1 Removal of the logarithmic factor

Eliminating the logarithmic term, replacing $(\beta/\beta_e)\cdot[\ln(8\beta)/\ln(8\beta_e)]$ with β/β_e, leads to dramatic degradation: the median error rises from 0.38% to 35.8% and the RMS error from 0.85% to 37.2%. High-β states are systematically underpredicted, and structured agreement within families is entirely lost.

A.2 Power-law deformation of the topological factor

Replacing the quadratic invariant with a general power, $(p^2+q^2) \to (p^2+q^2)^\alpha$, produces systematic distortions. Even modest deviations $|\alpha - 1| = 0.1$ increase the median error from 0.38% to $\sim$18–24%, with non-uniform misalignment across (p, q) sectors, confirming that the quadratic dependence is fixed by the effective circulation scaling.

A.3 Deformation of the confinement scaling

Modifying the confinement factor, $n_q^q \to n_q^{\gamma q}$, directly alters inter-class separation. Small deviations $|\gamma - 1| = 0.05$ raise the median error to $\sim$19–24%, while $|\gamma - 1| = 0.1$ increases it beyond 35%, reflecting the exponential sensitivity of the mass hierarchy to this factor.

A.4 Summary of sensitivity

Modification	Median	RMS	Max
Baseline formula	0.38%	0.85%	2.3%
No ln term	35.8%	37.2%	59.4%
$\alpha = 0.9$	18.4%	18.5%	28.5%
$\alpha = 1.1$	23.6%	23.7%	41.0%
$\gamma = 0.95$	19.3%	19.4%	29.5%
$\gamma = 1.05$	24.3%	25.1%	42.0%

Across all tested deformations, the mass spectrum exhibits extreme sensitivity to the functional form of each factor. Altering any component by as little as 5–10% leads to a roughly 50-fold increase in the median error relative to the baseline. This confirms that the structure of Eq. (10) is tightly constrained by the underlying vortex energetics, topology, and confinement mechanism, rather than representing a flexible empirical fit.